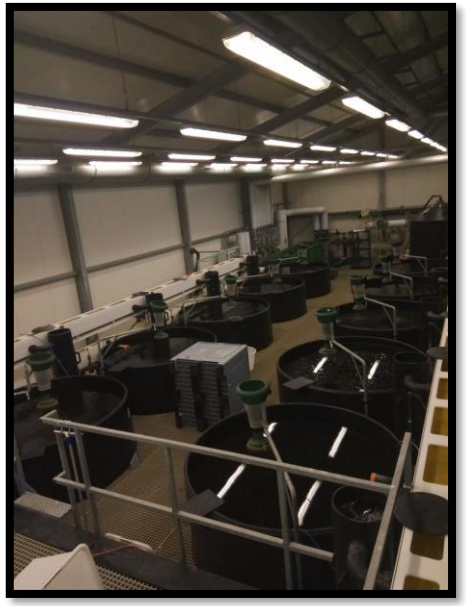
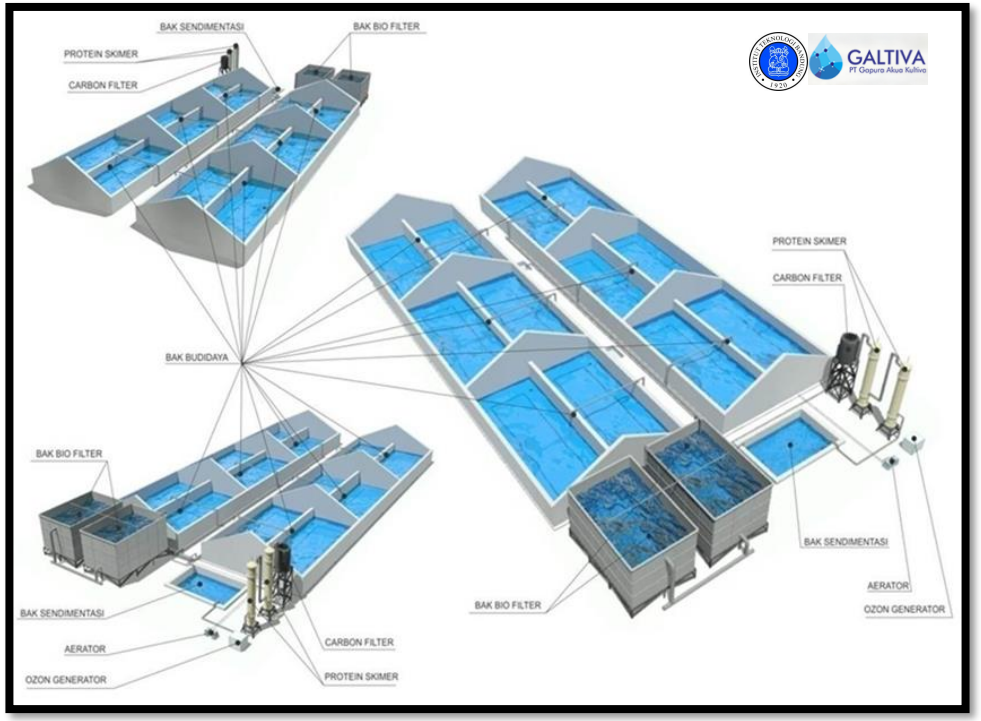
Green Technology













Closed System Aquaculture

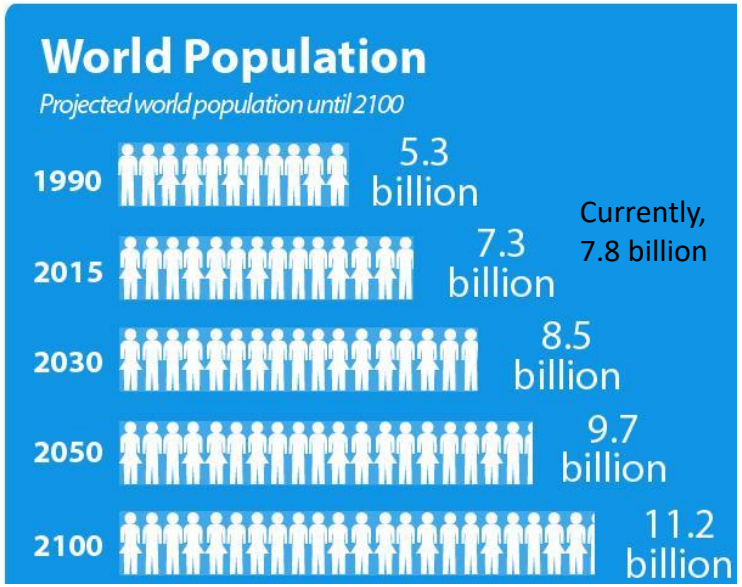
Harish Muhammad[1,2]
[1]M.Sc student at Humboldt University Berlin
[2]Dept. Ecophysiology and Aquaculture, Leibniz institute for Freshwater Ecology and Inland Fisheries

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"Pitching Nation 2020"

Challenge: feed the world in 2050!



UN Department of Economic & Social Affairs, 2015

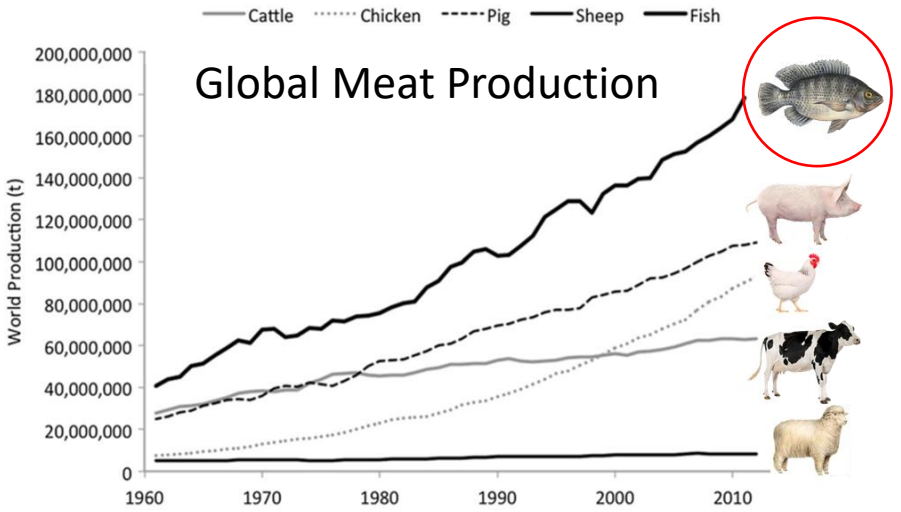
We have to feed 2 billion more people

But, shrinking resources, spaces, and nutrient

- Unsustainable exploitation
- Climate change

Aquaculture is the fastest-growing food sector

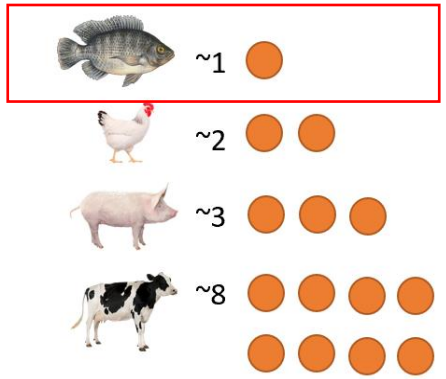
the production will be expected to double within 15-20 years (SOFIA FAO, 2018)



FAO stat 1960-2020

Protein production efficiency

Modified from De Silva & Anderson 1995

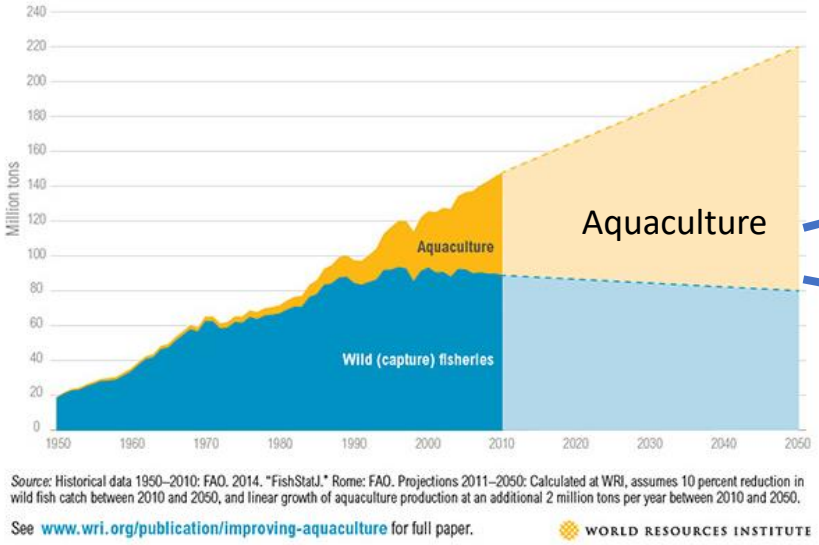


Fish is one of the most efficient live stocks

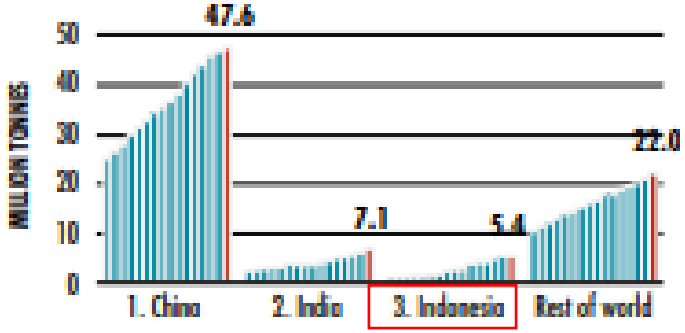
Kg feed needed to increase bodyweight by 1 kg

Contribution of Indonesia?

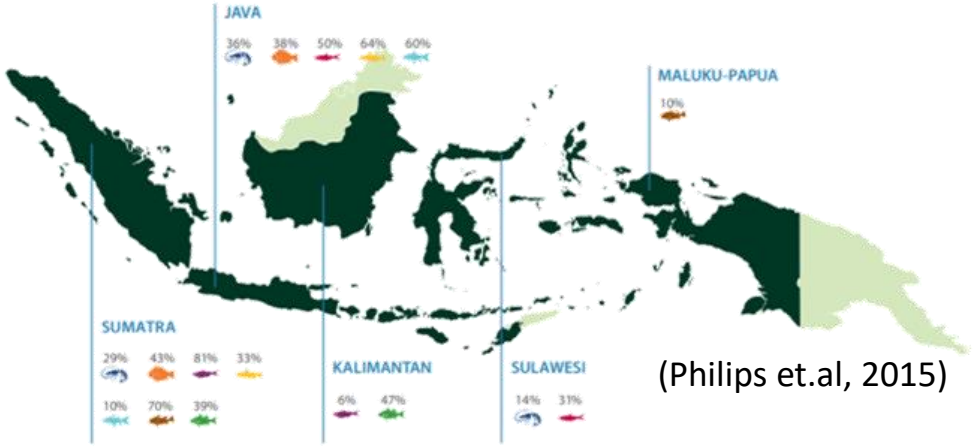
Global Fish Production



The 3rd largest aquaculture producer in the world



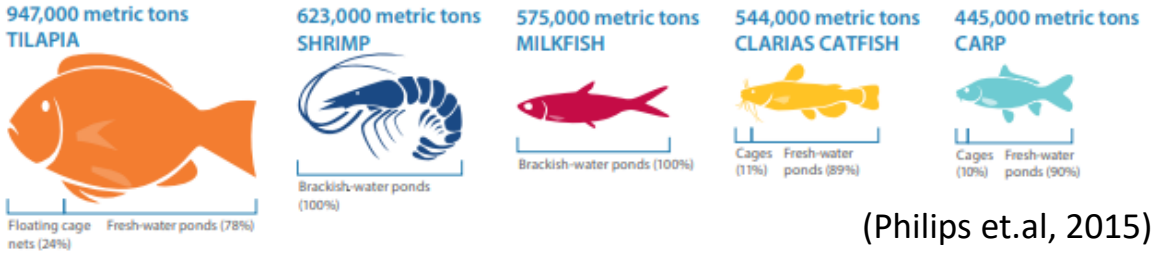
FAO, 2020



(Philips et.al, 2015)

Aquaculture produces 3.74 Millions job in Indonesia (KKP, 2017)

Indonesian's aquaculture commodities



(Philips et.al, 2015)

Existing Aquaculture Production in Indonesia

Open system leads to non sustainable production and less market acceptability

Sustainability



Intensive shrimp pond



Mangrove destruction

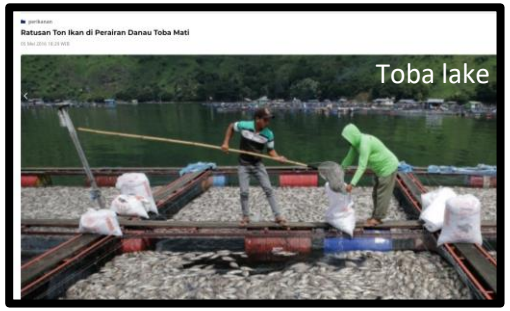


Pollution from effluent



Net cages for fish

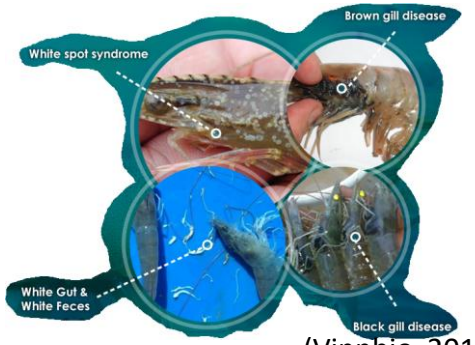
Point source pollution



Eutrophication



Excess use of antibiotics



(Vinnbio, 2017)

Emerging shrimp diseases

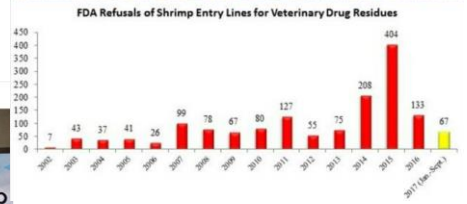
Market acceptability



Customer perceptions









Lets change our mindset in aquaculture from ‘open’ to ‘closed’ system!



Open system



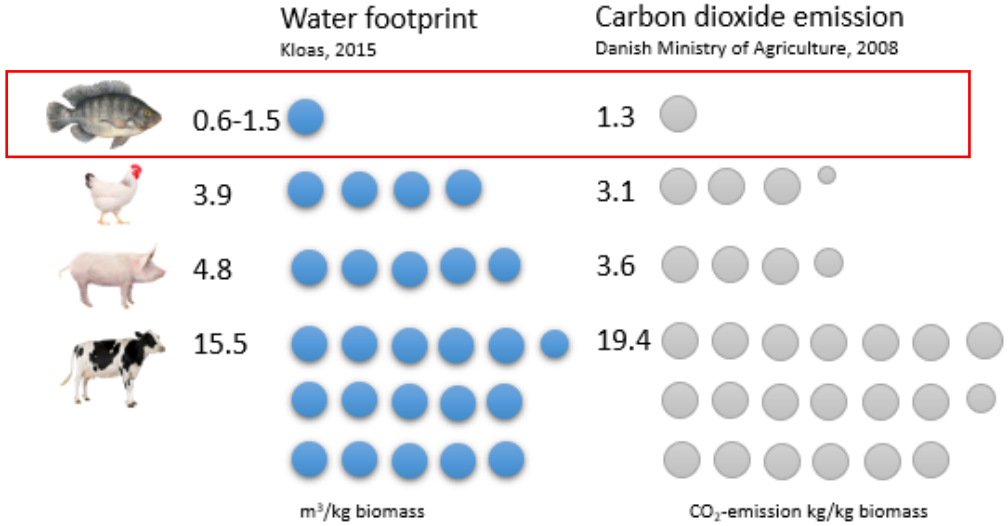
Closed system

More control

Supports higher density → higher productivity

Minimal environmental impact

Lower water footprint and CO_2 emission



Lower water consumption than conventional system

Type of system	Daily water use /day of total system water volume	Consumption of new water/kg fish produced	Degree of recirculation at system
Conventional Flow-through	1028%	30 m^3	0%
Closed system RAS	6%	0.3-3 m^3	95.9-99.6%

Bregnballe, 2015

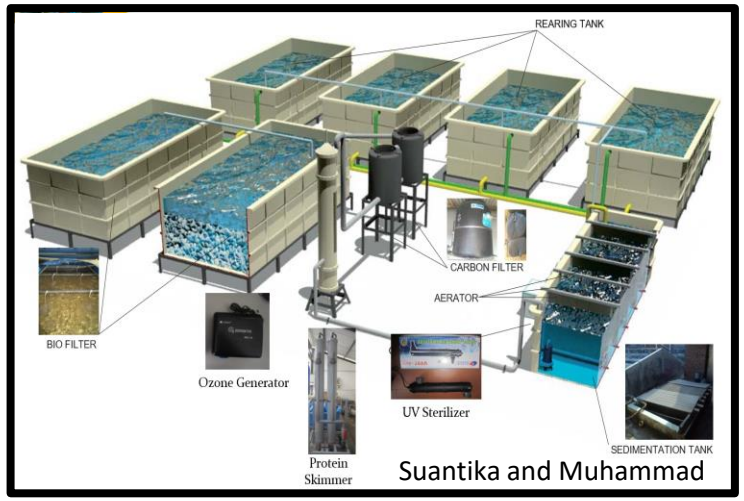
Better branding strategy

‘Clean product’ (no hormones, antibiotics)

‘Eco friendly’ potential to get ecolabelling

Sustainable

Example of closed system technology



Suantika and Muhammad







Recirculating Aquaculture System (RAS)

(+)Greater control
(+)Better water quality
(+)Low disease potential

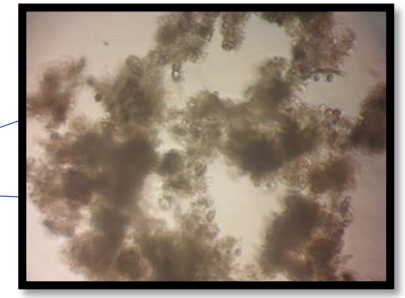
(-)More filter - Investment cost





Growth comparison


Suantika



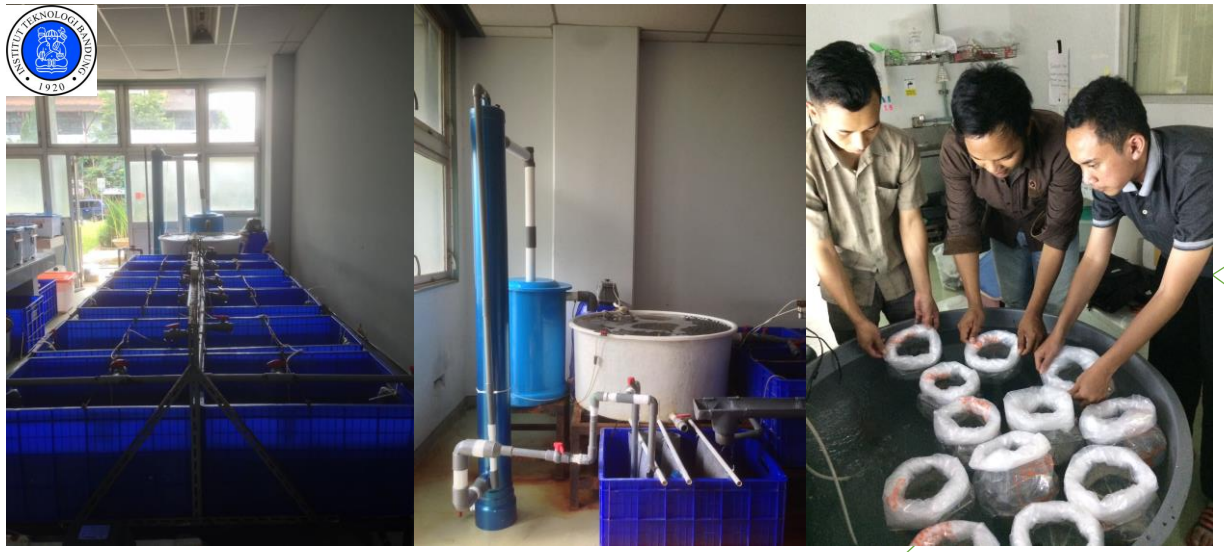
Zero Water Discharge (ZWD) and **Biofloc** system

(+)Lower feed quantity
(+)Less investment

(-)Less control
(-)More attention to water quality

‘Green impact’ for Indonesian fish farmer

From lab to semi-commercial application



ITB, Bandung

RAS Technology



RAS Technology



Hybrid RAS and ZWD



What can we do?

Be a wise consumer!!!

Show your concerns!



Our food is our choice!

Demand fresh and healthy food

Aquaculture industry is still young

Let’s push the industry in sustainable way!

Thank you and help me to promote GAIN!





For our future sustainable food production

Research for closed system aquaculture



Contact me:

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@harish_muh
Harish muhammad

GAIN
Green Aquaculture Indonesia

Thanks to:

lpdp
Lembaga pengelola dana pendidikan
RISTEKDIKTI